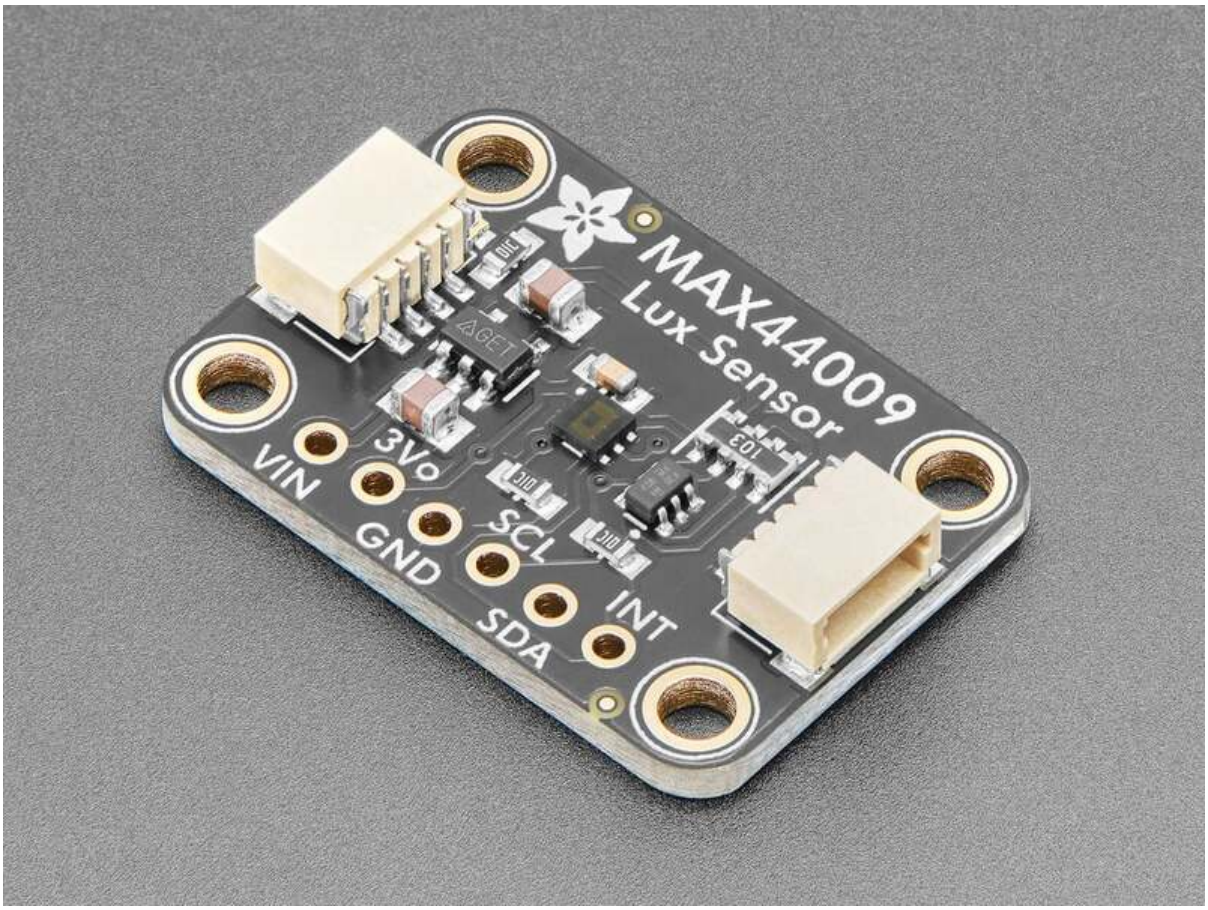
# Adafruit MAX44009 Lux Light Sensor

Created by Tim C




https://learn.adafruit.com/adafruit-max44009-lux-light-sensor

Last updated on 2026-05-07 04:24:27 PM UTC

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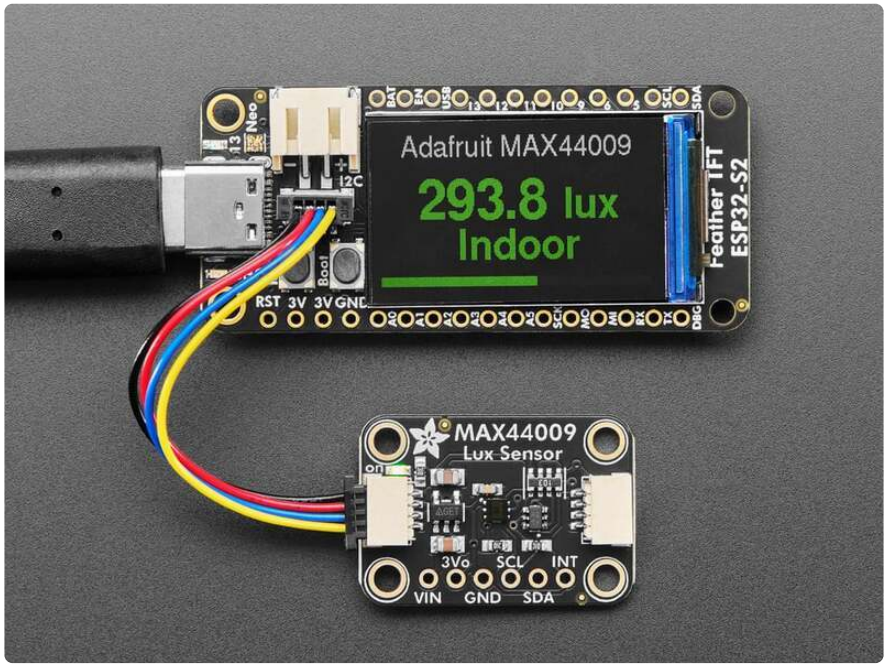
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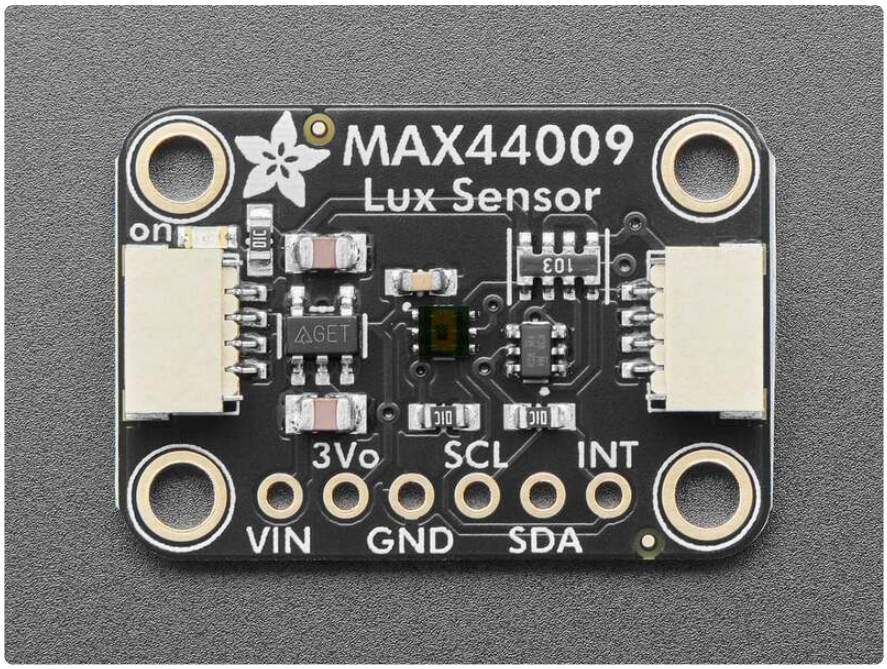
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# Overview



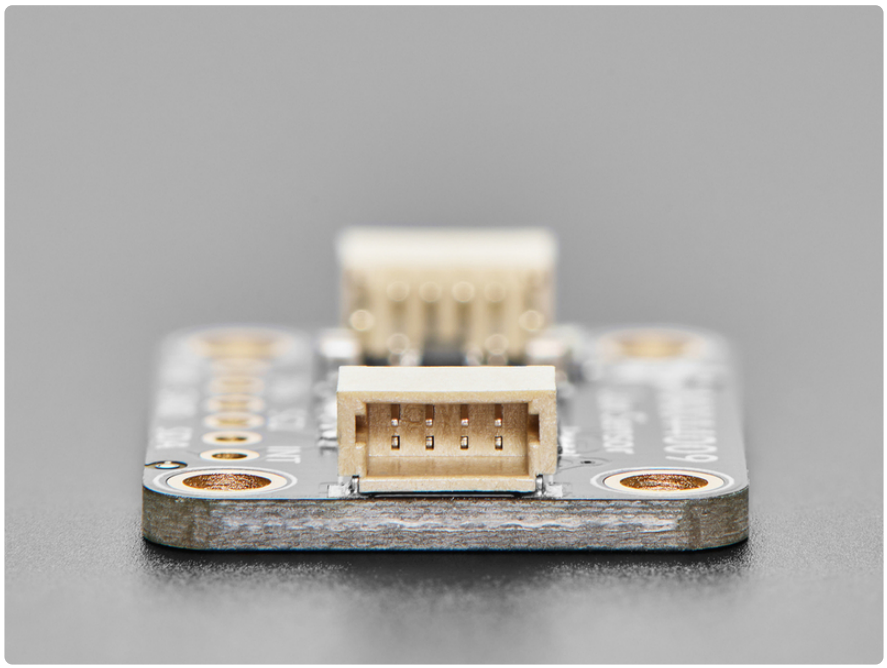

This is an easy to use light / lux sensor which features an ultra-wide 22-bit dynamic range from 0.045
lux to 188,000 lux. That means you can use the **Adafruit MAX44009 Wide-range Lux Sensor** in darkness or in bright outdoors sun, without having to tweak the integration time or gain: the sensor will auto-range for you so you get smooth readings no matter the light level.

Most light sensors just give you a number for brighter/darker ambient lighting. The MAX44009 makes your life easier by calculating the lux, which is an SI unit for light. You'll get more consistent readings between multiple sensors because you aren't dealing with some unit-less values.

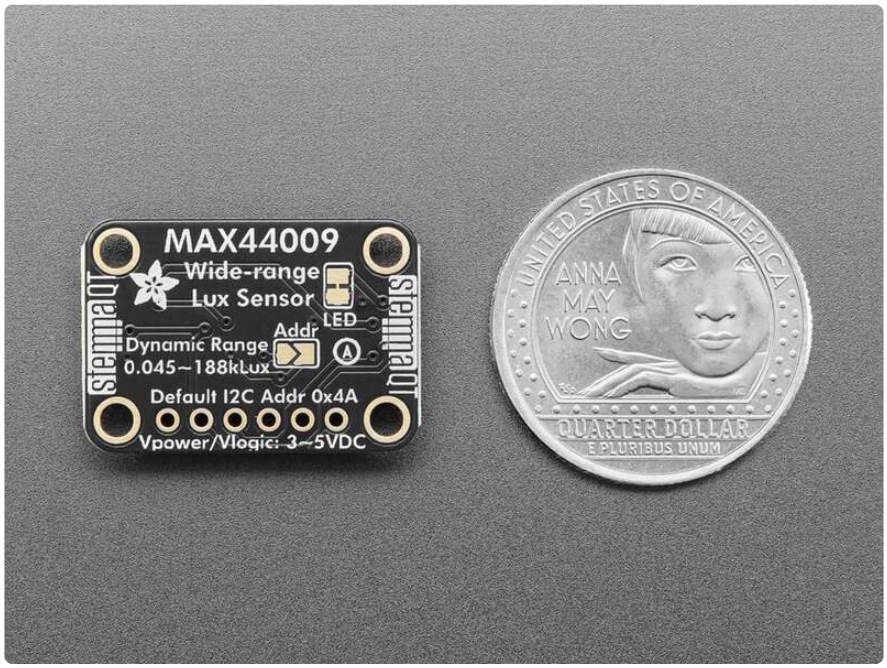



The sensor has 22-bit dynamic range for ambient light detection from 0.045 lux to about 188 klux with resolution down to 0.045 per least-significant-bit. It does come with software-adjustable integration times, or you can have it auto-range for you.

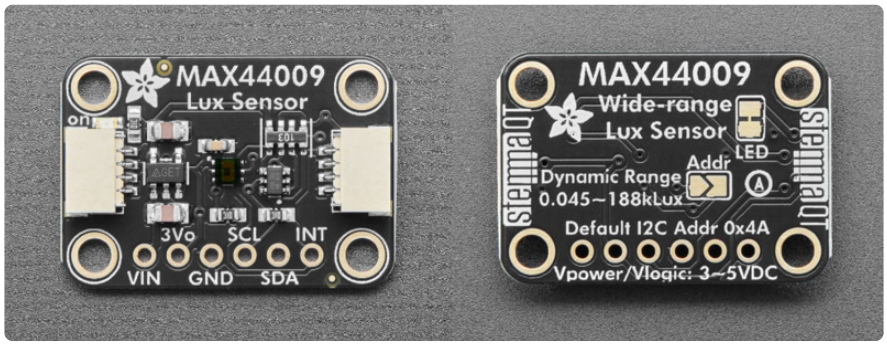
Interfacing is easy. This sensor uses plain, universal I2C. The sensor is on a breakout board with a 3.3V regulator and logic level shifter so you can use it with 3.3V or 5V power/logic microcontrollers. [We have written libraries for Arduino (C/C++) as well as CircuitPython (Python 3)](https://adafru.it/1aCE) (https://adafru.it/1aCE)so you can use this sensor with just about any kind of device, even a Raspberry Pi!



As if that weren't enough, we've also added SparkFun qwiic (https://adafru.it/Fpw) compatible STEMMA QT **(https://adafru.it/Ft4)** connectors for the I2C bus **so you don't even need to solder.** Just wire up to your favorite micro with a plug-and-play cable to get lux data ASAP. For a no-solder experience, just wire up to your favorite micro, like the ESP32 V2 Feather (http://adafru.it/5400) using a STEMMA QT adapter cable (https://adafru.it/JnB). The Stemma QT connectors also mean the MAX44009 can be used with our various associated accessories (https://adafru.it/Ft6). QT Cable is not included, but we have a variety in the shop (https://adafru.it/17VE).

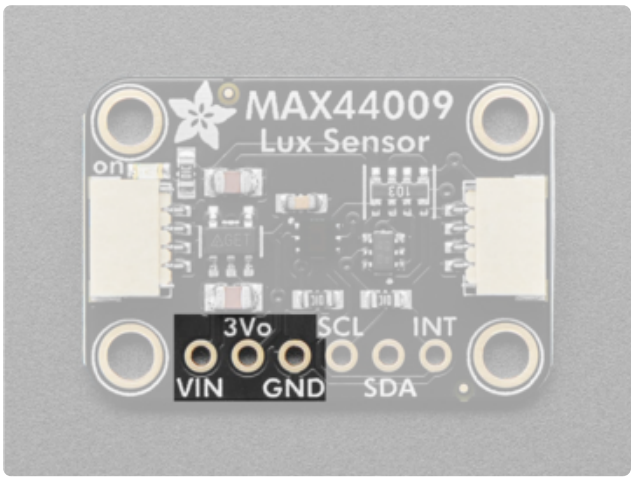



# Pinouts



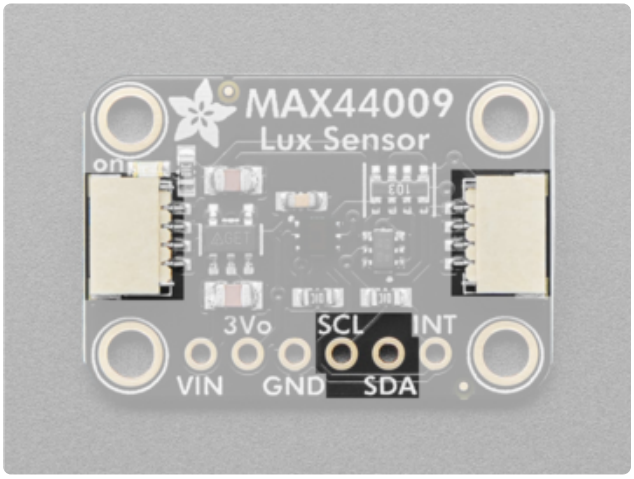

The default I2C address is **0x4A**.

## Power Pins



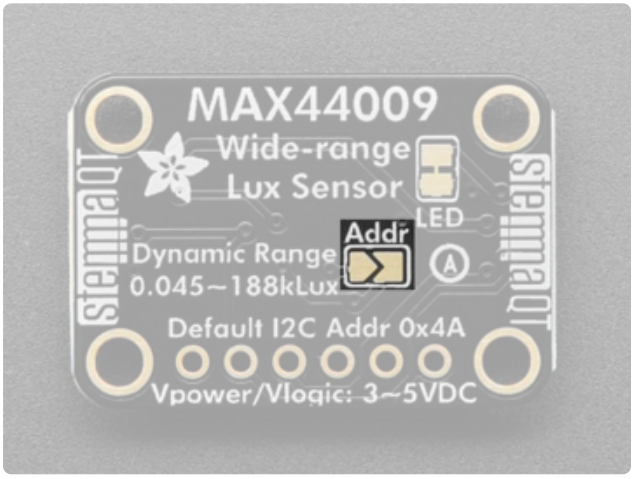

**Vin** - this is the power pin. Since the sensor chip uses 3VDC, we have included a voltage regulator on board that will take 3-5VDC and safely convert it to 3VDC. To power the board, give it the same power as the logic level of your microcontroller - e.g. for a 5V micro like Arduino, use 5V.
**3Vo** - this is the 3.3V output from the voltage regulator, you can grab up to 100mA from this if you like
**GND** - common ground for power and logic

## I2C Data Pins




**SCL** - the I2C clock pin, connect to your microcontroller I2C clock line.
**SDA** - the I2C data pin, connect to your microcontroller I2C data line.
STEMMA QT **(https://adafru.it/Ft4) -** These connectors allow you to connectors to dev boards with **STEMMA QT** connectors or to other things with various associated accessories (https://adafru.it/Ft6)

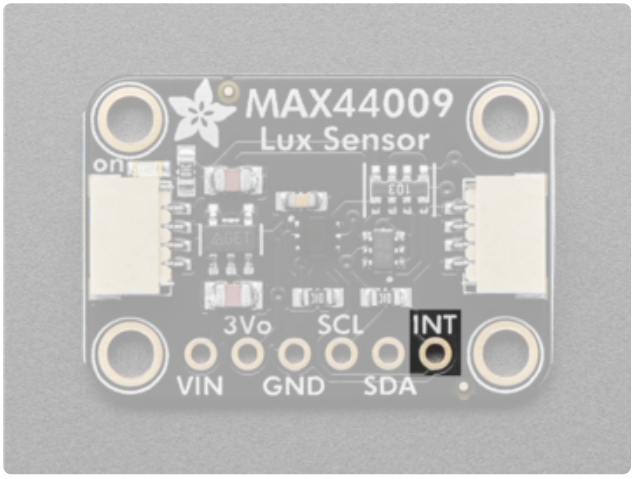
## I2C Address Select

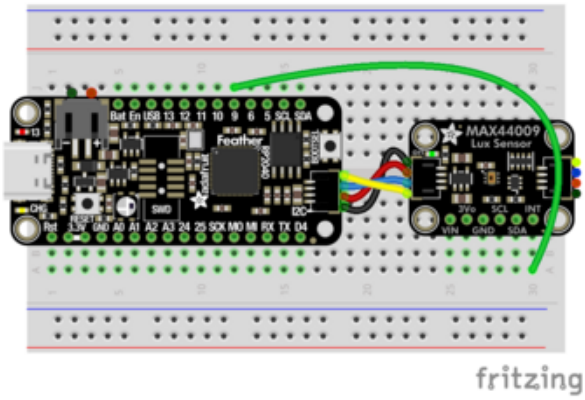



By default the I2C address on the MAX44009 breakout will be **0x4A**. An alternative address of **0x4B** is available by soldering closed the **Addr** jumper on the bottom of the breakout.

| ADDR | Connection | Jumper |
|---|---|---|
| 0x4A | GND | Open |
| 0x4B | VCC | Soldered |

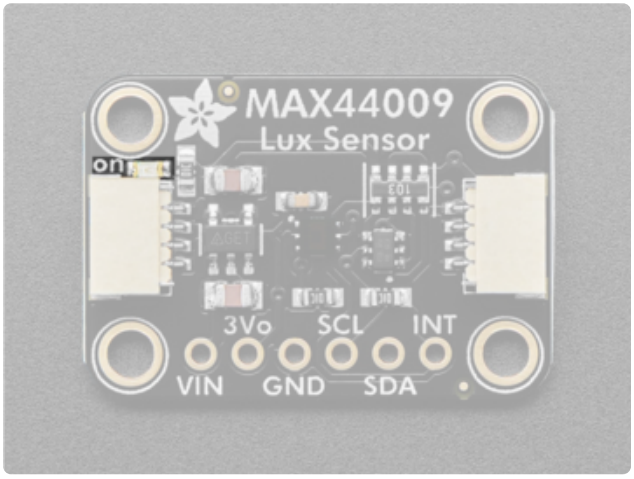
## Interrupt Pin



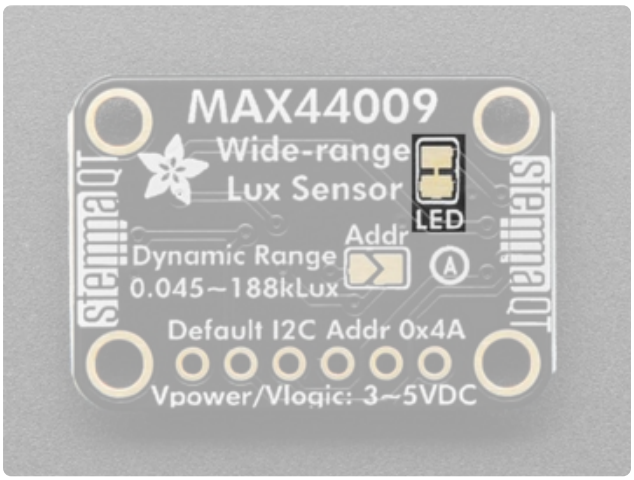




**INT** - the interrupt pin. It is an open-drain interrupt. To use, connect the pin to a GPIO pin on your microcontroller and configure as pull up.

## LED and LED Jumper




Above the left STEMMA QT connector there is a small green **ON** power indicator LED. When the breakout is powered this LED will turn on.




On the back of the MAX44009 breakout there is a jumper with a small trace between two pads that can be cut to disable the **ON** power indicator LED on the board.

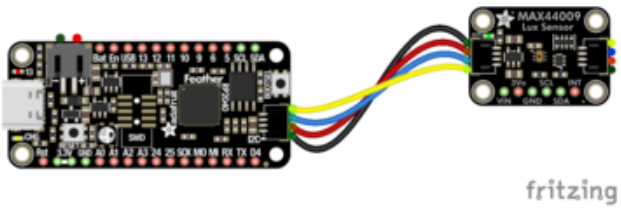
# CircuitPython and Python

It's easy to use the **MAX44009** with Python or CircuitPython, and the Adafruit_CircuitPython_MAX44009 (https://adafru.it/1aCF) module. This module allows you to easily write Python code to read data from the sensor.

You can use this driver with any CircuitPython microcontroller board or with a computer that has GPIO and Python thanks to Adafruit_Blinka, our CircuitPython-for-Python compatibility library (https://adafru.it/BSN).

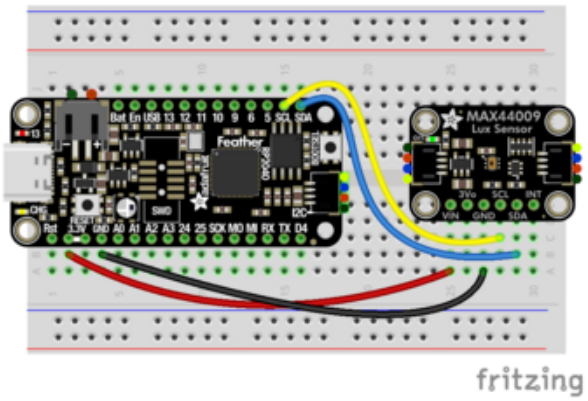
## CircuitPython Microcontroller Wiring

First wire up a MAX44009 to your board exactly as shown below. Here's an example of wiring a Feather RP2040 to the sensor with I2C using one of the handy **STEMMA QT** (https://adafru.it/Ft4) connectors:




**Board STEMMA 3V** to **sensor VIN (red wire)**
**Board STEMMA GND** to **sensor GND (black wire)**
**Board STEMMA SCL** to **sensor SCL (yellow wire)**
**Board STEMMA SDA** to **sensor SDA (blue wire)**

The following is the MAX44009 wired to a Feather RP2040 using a solderless breadboard:

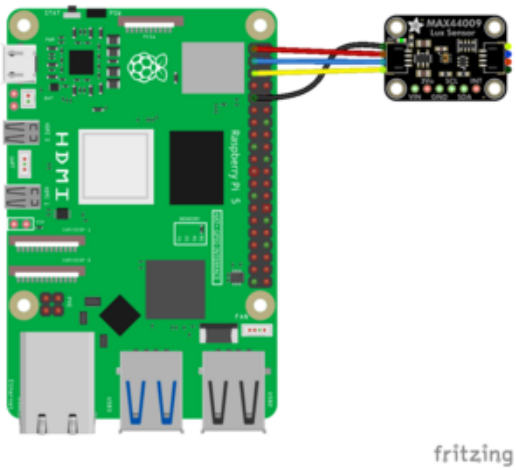



**Board 3V** to **sensor VIN (red wire)**
**Board GND** to **sensor GND (black wire)**
**Board SCL** to **sensor SCL (yellow wire)**
**Board SDA** to **sensor SDA (blue wire)**

## Python Computer Wiring

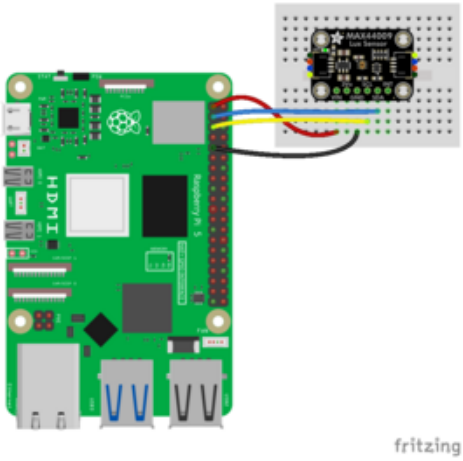
Since there are dozens of Linux computers/boards you can use, we will show wiring for Raspberry Pi. For other platforms, [please visit the guide for CircuitPython on Linux to see whether your platform is supported](https://adafru.it/BSN) (https://adafru.it/BSN).

Here's the Raspberry Pi wired with I2C using the STEMMA connector:




**Pi 3V** to **sensor VIN (red wire)**
**Pi GND** to **sensor GND (black wire)**
**Pi SCL** to **sensor SCL (yellow wire)**
**Pi SDA** to **sensor SDA (blue wire)**

Here's the Raspberry Pi wired with I2C using a solderless breadboard:




**Pi 3V** to **sensor VIN (red wire)**
**Pi GND** to **sensor GND (black wire)**
**Pi SCL** to **sensor SCL (yellow wire)**
**Pi SDA** to **sensor SDA (blue wire)**

## Python Installation of MAX44009 Library

You'll need to install the **Adafruit_Blinka** library that provides the CircuitPython support in Python. This may also require enabling I2C on your platform and verifying you are running Python 3. Since each platform is a little different, and Linux changes often, please visit the CircuitPython on Linux guide to get your computer ready (https://adafru.it/BSN)!

Once that's done, from your command line run the following command:

- `pip3 install adafruit-circuitpython-max44009`

If your default Python is version 3 you may need to run 'pip' instead. Just make sure you aren't trying to use CircuitPython on Python 2.x, it isn't supported!

## CircuitPython Usage

To use with CircuitPython, you need to first install the **Adafruit_CircuitPython_MAX44009** library, and its dependencies, into the **lib** folder on your **CIRCUITPY** drive. Then you need to update **code.py** with the example script.

Thankfully, we can do this in one go. In the example below, click the **Download Project Bundle** button below to download the necessary libraries and the **code.py** file in a zip file. Extract the contents of the zip file, and copy the **entire lib folder** and the **code.py** file to your **CIRCUITPY** drive.

Your **CIRCUITPY/lib** folder should contain the following folders and file:

- **adafruit_bus_device/**
- **adafruit_register/**
- **adafruit_max44009.mpy**




## Python Usage

Once you have the library `pip3` installed on your computer, copy or download the following example to your computer, and run the following, replacing **code.py** with whatever you named the file:

```
python3 code.py
```

## Example Code

**If running CircuitPython:** Once everything is saved to the **CIRCUITPY** drive, [connect to the serial console](https://adafru.it/Bec) (https://adafru.it/Bec) to see the data printed out!

**If running Python:** The console output will appear wherever you are running Python.

```
# SPDX-FileCopyrightText: Copyright (c) 2026 Tim Cocks for Adafruit Industries
#
# SPDX-License-Identifier: MIT
"""
Adafruit MAX44009 Simple Test

Basic example for the Adafruit MAX44009 library.
Reads ambient light lux values.
"""

import time

import board

from adafruit_max44009 import MAX44009

print("Adafruit MAX44009 Simple Test")

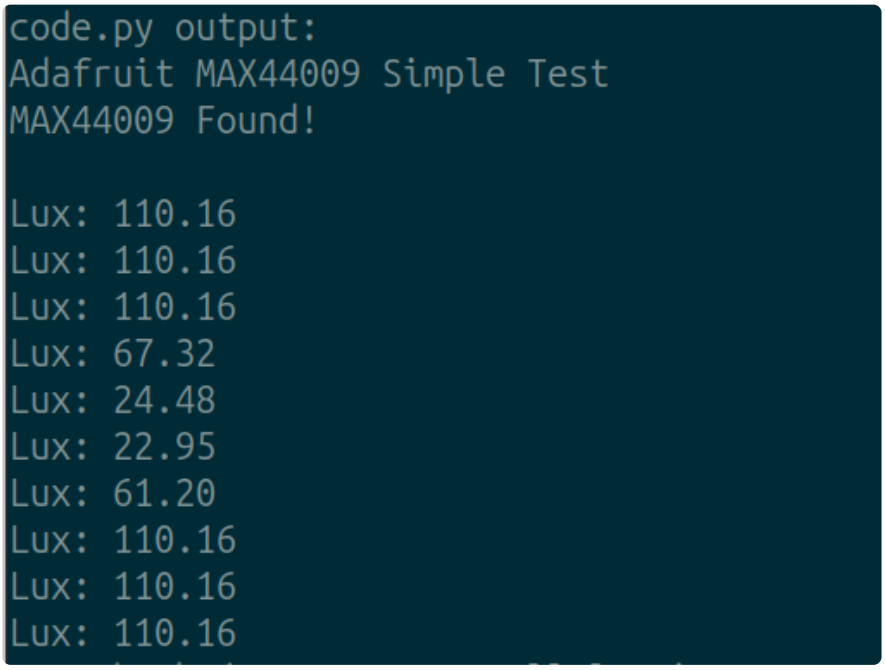
sensor = MAX44009(board.I2C())

print("MAX44009 Found!")
print()

while True:
    print(f"Lux: {sensor.lux:.2f}")
    time.sleep(1.0)
```

https://github.com/adafruit/Adafruit_CircuitPython_MAX44009/blob/main/examples/max44009_simpletest.py

First, the sensor gets initialized over I2C. In the main loop the lux data is read from the sensor and the values are printed to the serial console once per second.



# Python Docs

Python Docs (https://adafru.it/1aCA)

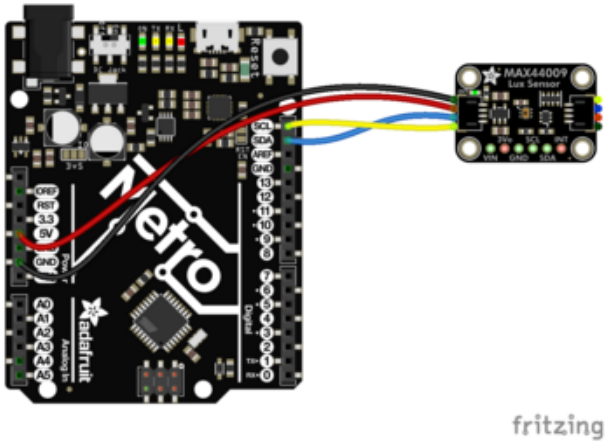
# Arduino

Using the MAX44009 breakout with Arduino involves wiring up the breakout to your Arduino-compatible microcontroller, installing the Adafruit_MAX44009 (https://adafru.it/1aCG) library, and running the provided example code.

## Wiring

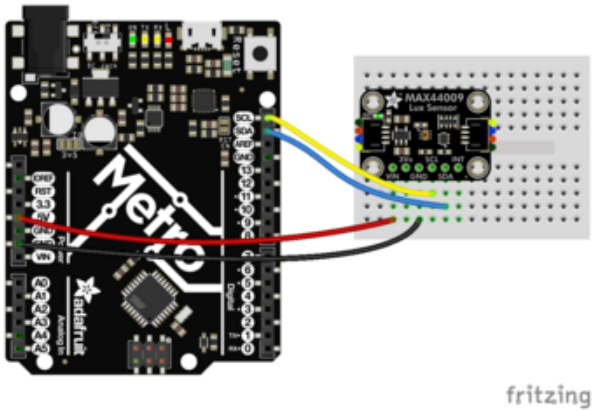
Wire as shown for a **5V** board like an Uno. If you are using a **3V** board, like an Adafruit Feather, wire the board's 3V pin to the sensor **VIN**.

Here is an Adafruit Metro wired up to the sensor using the STEMMA QT connector:




**Board 5V** to **breakout VIN (red wire)**
**Board GND** to **breakout GND (black wire)**
**Board SCL** to **breakout SCL (yellow wire)**
**Board SDA** to **breakout SDA (blue wire)**

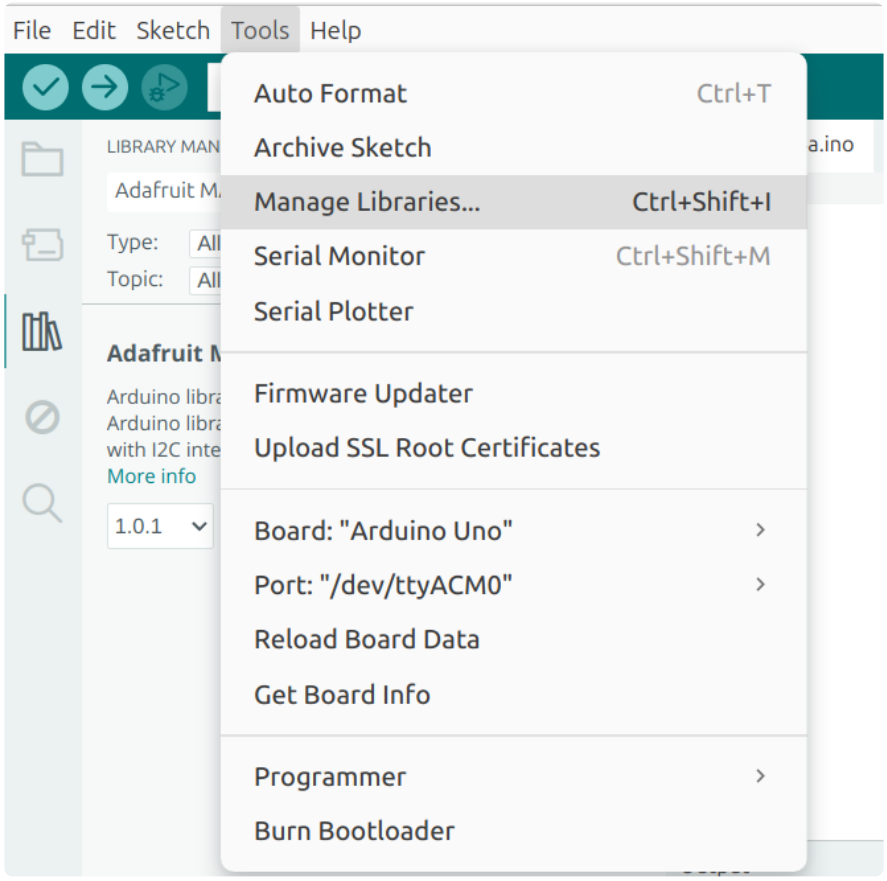
Here is an Adafruit Metro wired up using a solderless breadboard:



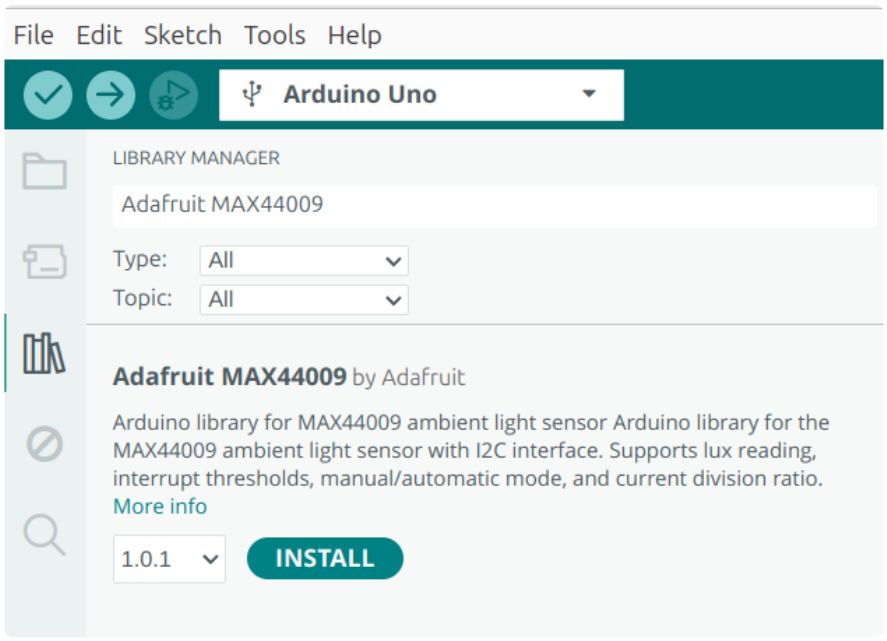

**Board 5V** to **breakout VIN (red wire)**
**Board GND** to **breakout GND (black wire)**
**Board SCL** to **breakout SCL (yellow wire)**
**Board SDA** to **breakout SDA (blue wire)**

## Library Installation

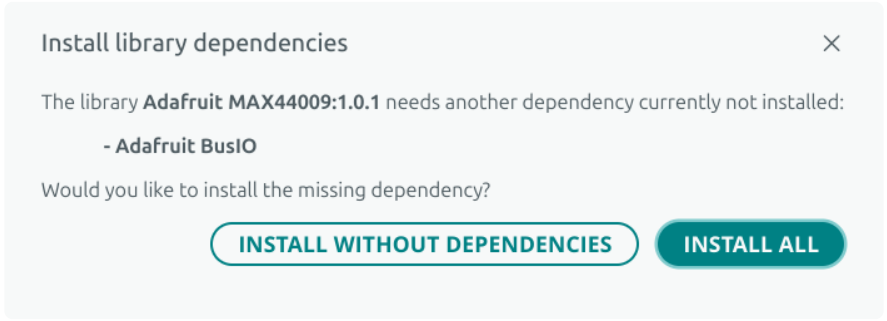
You can install the **Adafruit MAX44009** library for Arduino using the Library Manager in the Arduino IDE.




Click the **Manage Libraries ...** menu item, search for **Adafruit MAX44009**, and select the **Adafruit MAX44009** library:




If asked about dependencies, click "Install all".




If the "Dependencies" window does not come up, then you already have the dependencies installed.

If the dependencies are already installed, you must make sure you update them through the Arduino Library Manager before loading the example!

## Example Code

```
1  /*
2   * Simple example for the MAX44009 Ambient Light Sensor
```

```
 *
 * Written by Limor 'ladyada' Fried with assistance from Claude Code for
 * Adafruit Industries. MIT license, check license.txt for more information
 *
 * Reads lux every 100ms in auto-ranging mode. Serial Plotter friendly.
 */

#include <Adafruit_MAX44009.h>

Adafruit_MAX44009 max44009;

void setup() {
  Serial.begin(115200);
  while (!Serial)
    delay(10);

  if (!max44009.begin()) {
    Serial.println(F("Could not find MAX44009 sensor!"));
    while (1)
      delay(10);
  }
}

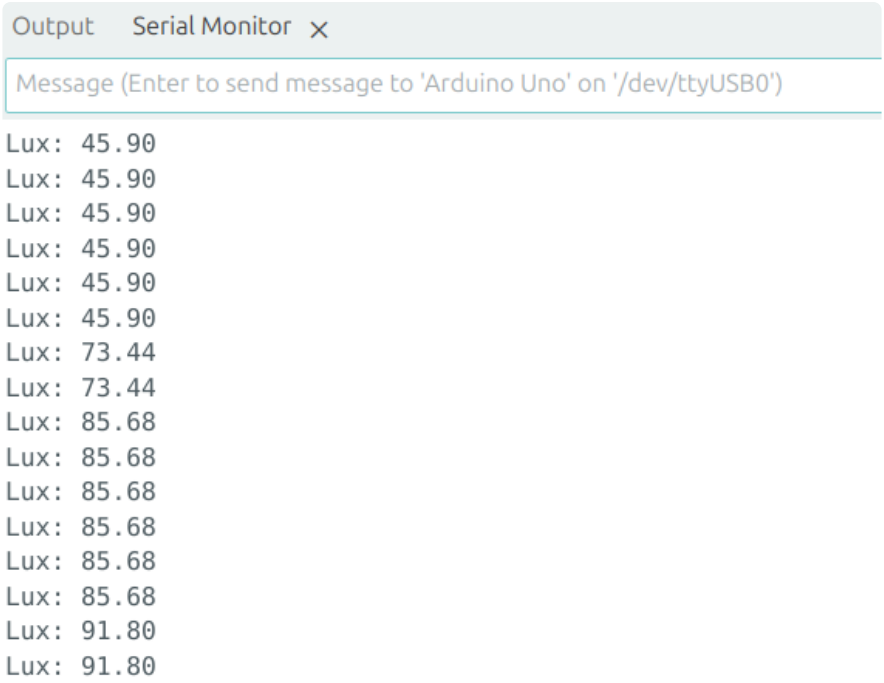
void loop() {
  float lux = max44009.readLux();

  if (!isnan(lux)) {
    Serial.print(F("Lux: "));
    Serial.println(lux);
  }

  delay(100);
}
```

https://github.com/adafruit/Adafruit_MAX44009/blob/main/examples/max44009_simpletest/max44009_simpletest.ino

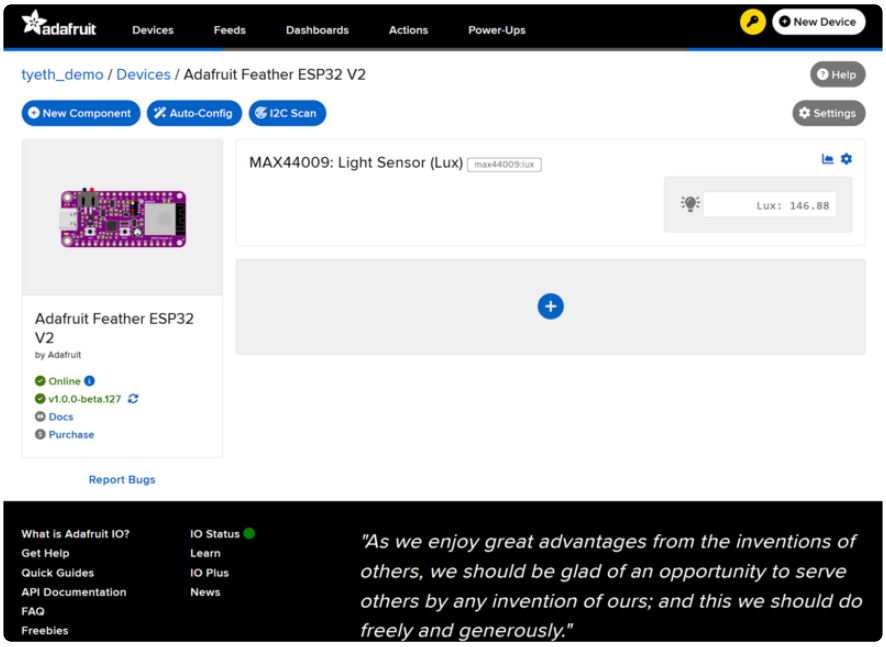
Upload the sketch to your board and open up the Serial Monitor (**Tools -> Serial Monitor**) at 115200 baud. You'll see the MAX44009 recognized over I2C. Then, the reading values of different light channels will be printed out to the Serial Monitor.




# Arduino Docs

[Arduino Docs](https://adafru.it/1aCB) (https://adafru.it/1aCB)

# WipperSnapper



## What is WipperSnapper

WipperSnapper is a firmware designed to turn any WiFi-capable board into an Internet-of-Things device without programming a single line of code. WipperSnapper connects to Adafruit IO (https://adafru.it/fsU), a web platform designed (by Adafruit! (https://adafru.it/Bo5)) to display, respond, and interact with your project's data.

Simply load the WipperSnapper firmware onto your board, add credentials, and plug it into power. Your board will automatically register itself with your Adafruit IO account.

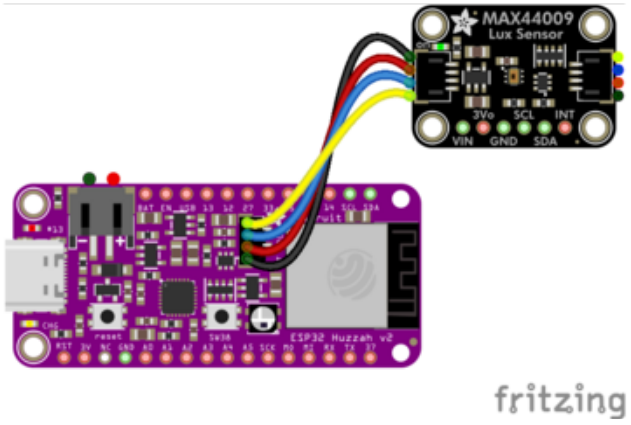
From there, you can add components to your board such as buttons, switches, potentiometers, sensors, and more! Components are dynamically added to hardware, so you can immediately start interacting, logging, and streaming the data your projects produce without writing code.

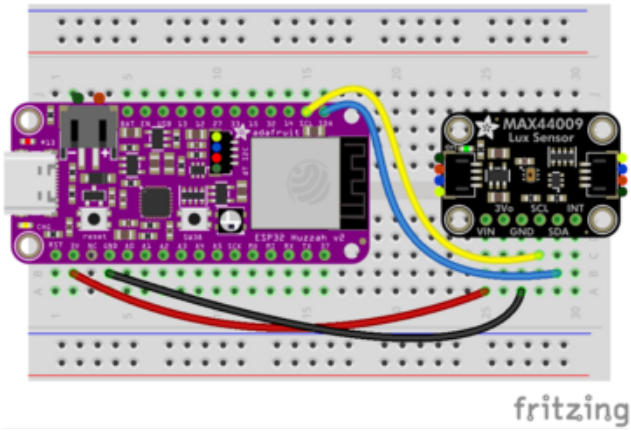
If you've never used WipperSnapper, click below to read through the quick start guide before continuing.

https://adafru.it/Vfd

## Wiring

Wire up the MAX44009 exactly as follows. Here it's shown connected via our convenient StemmaQT cable, or alternatively wired up on a prototyping breadboard:



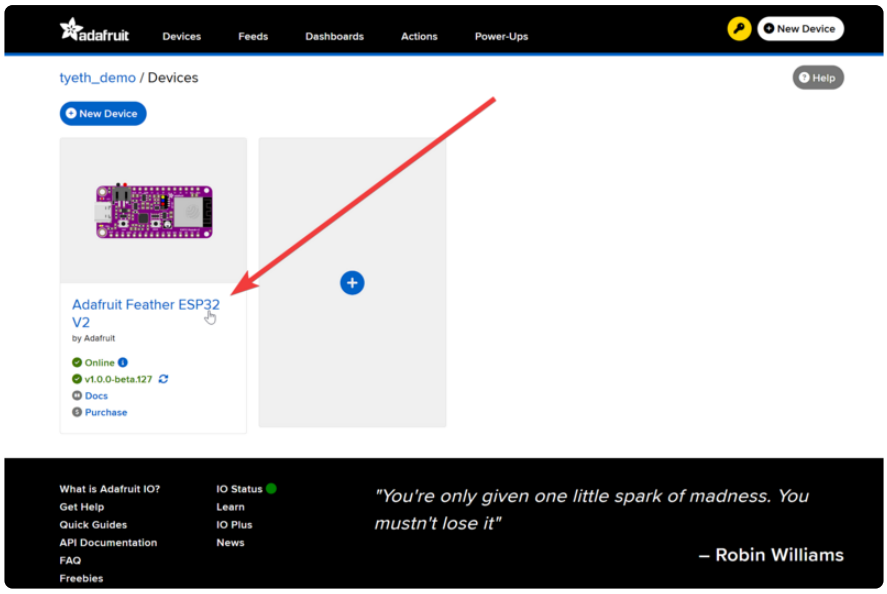




**Board 5V** to **breakout STEMMA VIN (red wire)**
**Board GND** to **breakout STEMMA GND (black wire)**
**Board SCL** to **breakout STEMMA SCL (yellow wire)**
**Board SDA** to **breakout STEMMA SDA (blue wire)**

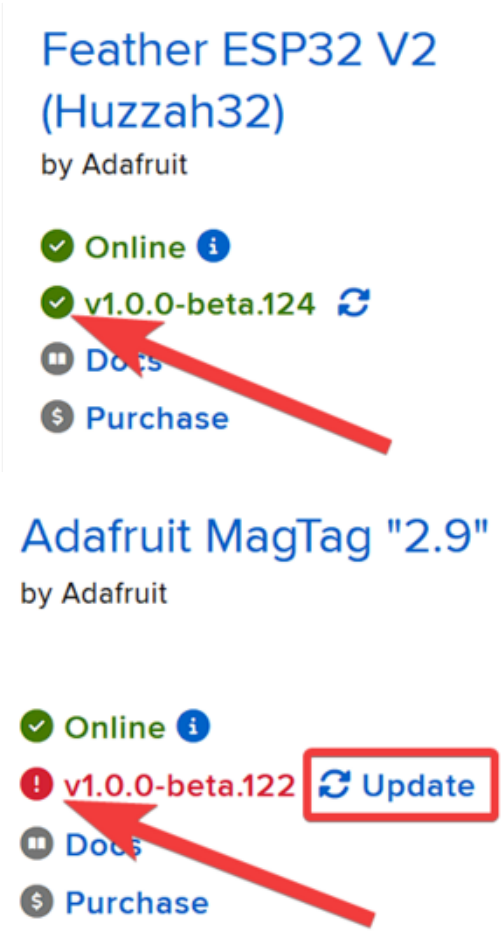
## Usage

Connect your board to Adafruit IO Wippersnapper and navigate to the WipperSnapper board list **(https://adafru.it/TAu)**.

On this page, **select the WipperSnapper board you're using** to be brought to the board's interface page.




If you do not see your board listed here - you need [to connect your board to Adafruit IO (https://adafru.it/Vfd)](https://adafru.it/Vfd) first.

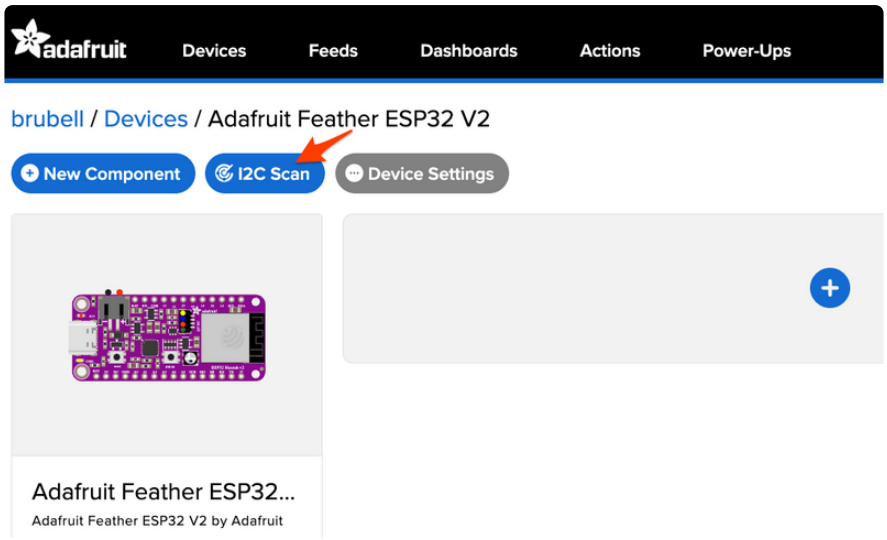



On the device page, quickly **check that you're running the latest version of the WipperSnapper firmware**.

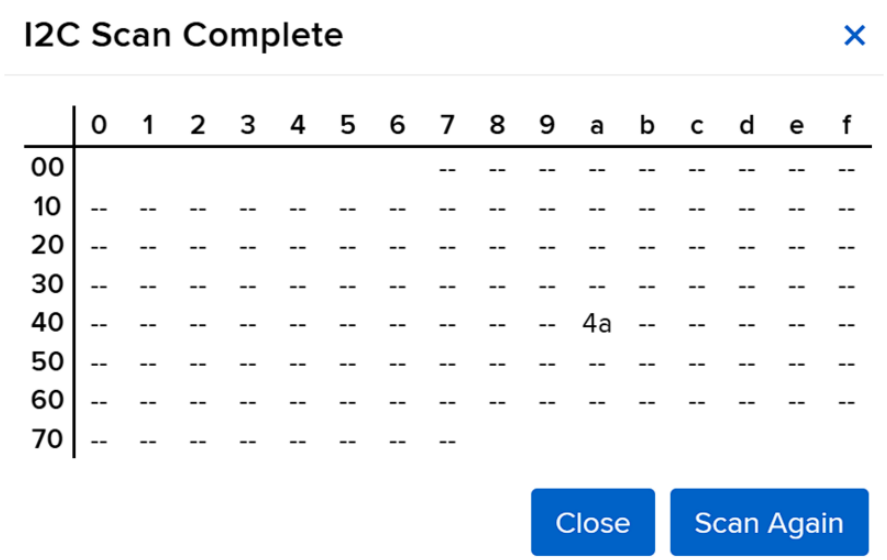
The device tile on the left indicates the version number of the firmware running on the connected board.

**If the firmware version is green with a checkmark -** continue with this guide.
**If the firmware version is red with an exclamation mark "!" -** [update to the latest WipperSnapper firmware (https://adafru.it/Vfd)](https://adafru.it/Vfd) on your board before continuing.

Next, make sure the sensor is plugged into your board and click the **I2C Scan** button.




You should see the MAX44009's default I2C address of `0x4a` pop-up in the I2C scan list.




| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | a | b | c | d | e | f |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 00 | | | | | | | | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| 10 | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| 20 | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| 30 | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| 40 | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | 4a | -- | -- | -- | -- | -- |
| 50 | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| 60 | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| 70 | -- | -- | -- | -- | -- | -- | -- | -- | | | | | | | | |

## I don't see the sensor's I2C address listed!

First, double-check the connection and/or wiring between the sensor and the board.

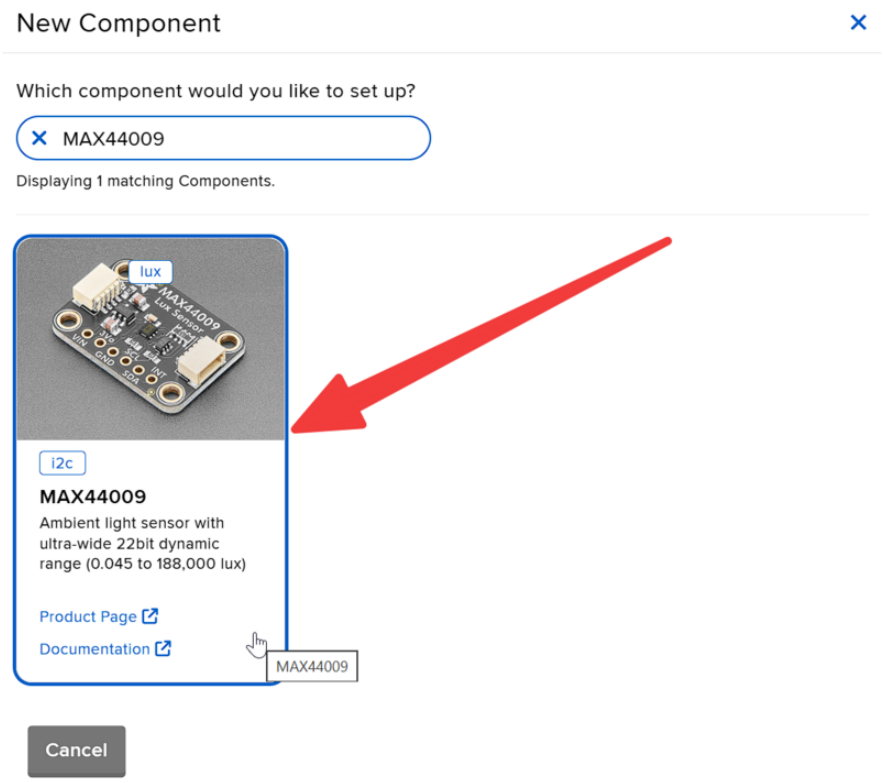
Then, reset the board and let it re-connect to Adafruit IO WipperSnapper.

With the sensor detected in an I2C scan, you're ready to add the sensor to your board.

**Click the New Component button or the + button** to bring up the component picker.



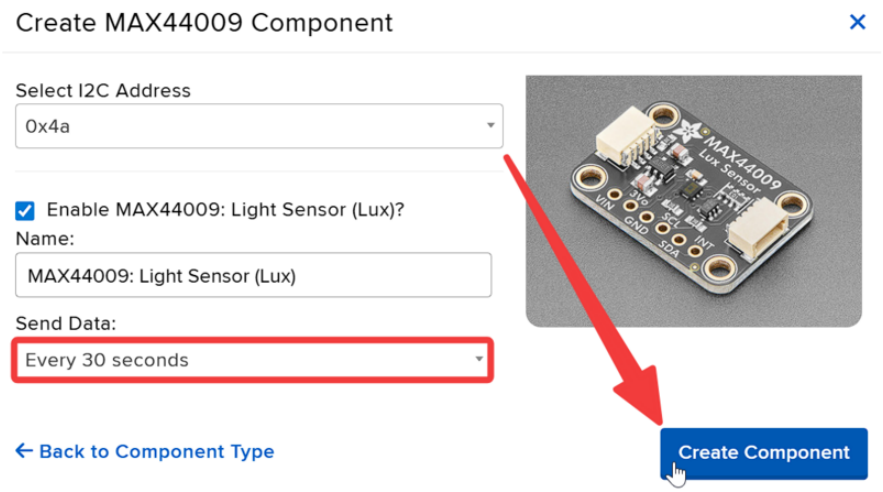

Adafruit IO supports a large amount of components. To quickly find your sensor, type `MAX44009` into the search bar, then select the **MAX44009** component.



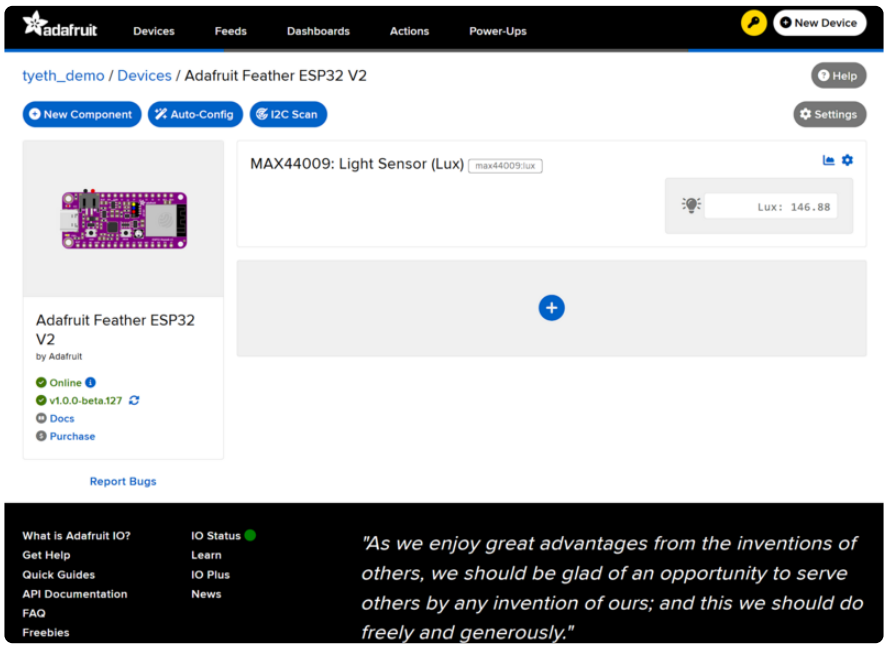

On the component configuration page, the MAX44009's sensor address should be listed along with the sensor's settings.

The **Send Every** option is specific to each sensor's measurements. This option will tell the Feather how often it should read from the MAX44009 sensor and send the data to Adafruit IO. Measurements can range from every second to every 24 hours.

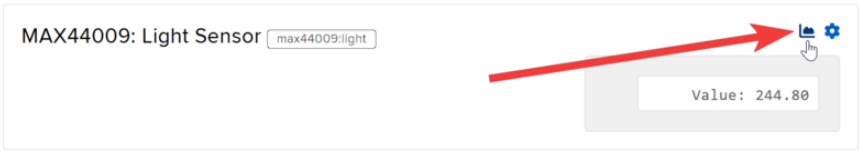
For this example, set the **Send Every** interval to every 30 seconds.



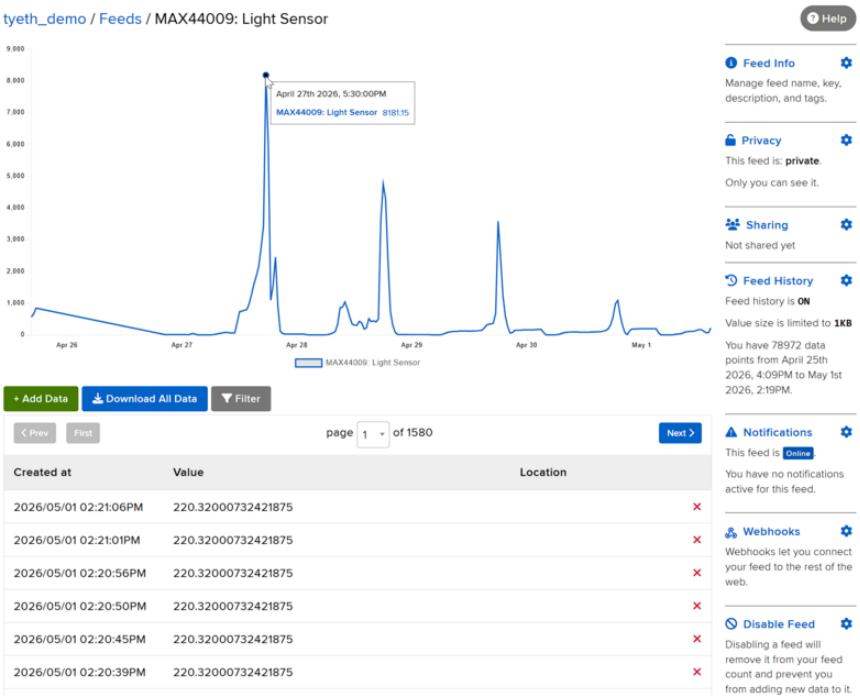
Your device interface should now show the sensor components you created. After the interval you configured elapses, WipperSnapper will automatically read values from the sensor(s) and send them to Adafruit IO.




To view the data that has been logged from the sensor, click on the graph next to the sensor name.




Here you can see the feed history and edit things about the feed such as the name, privacy, webhooks associated with the feed and more. If you want to learn more about how feeds work, check out this page (https://adafru.it/10aZ).

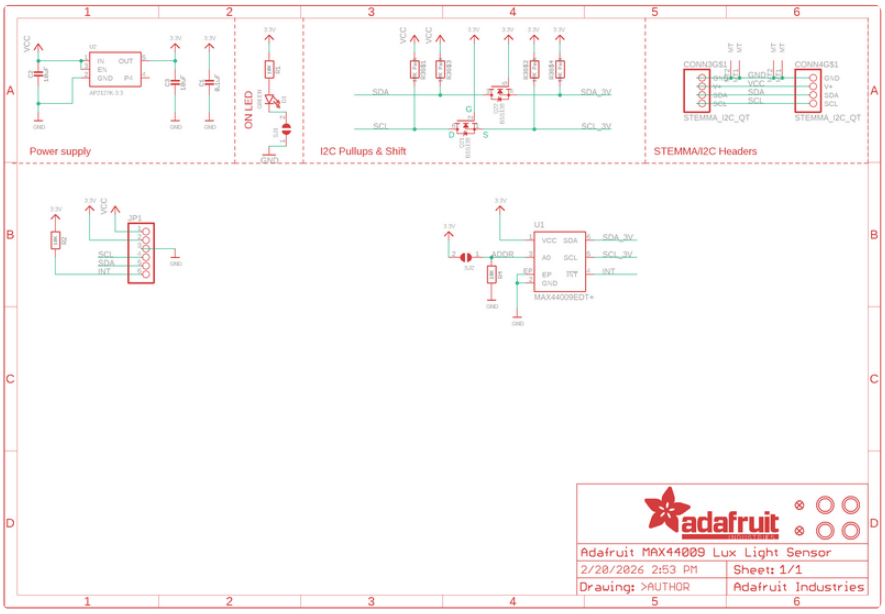


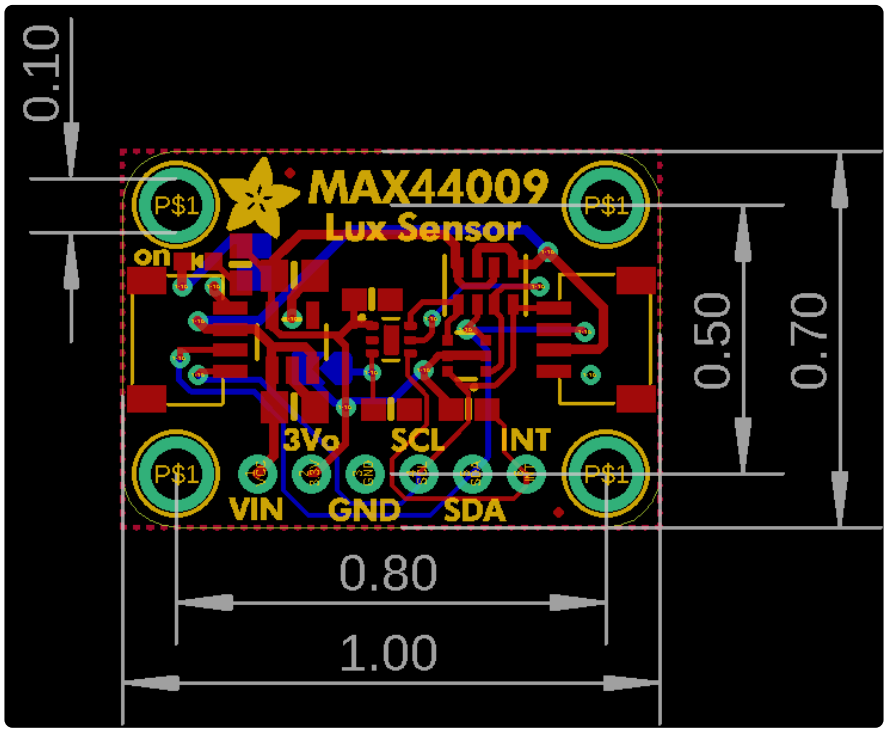

# Downloads

## Files

- MAX44009 Datasheet (https://adafru.it/1aCH)
- EagleCAD PCB files on GitHub (https://adafru.it/1aCI)
- 3D models on GitHub (https://adafru.it/1aCP)
- Fritzing object in the Adafruit Fritzing Library (https://adafru.it/1aCJ)

# Schematic and Fab Print



## 3D Model

